

Australian SMOG Model Coefficient and Fit Analysis

Model output review

16 September 2026
Saved model results through 2026Q1

1 Scope and interpretation

This report analyzes the standalone Australian Python SMOG model. Estimates come from its saved parameter JSON and accepted-iteration likelihood history. The generation sample is 1980Q1–2023Q4 (176 quarters). The update applies those fixed coefficients to later data; the latest complete saved result is 2026Q1. The latent potential output and NAIRU are model estimates, not directly observed quantities.

2 Results and charts

The fixed-parameter update runs through 2026Q1. Its saved smoothed output gap is -0.371% of potential output and its NAIRU is 4.089% . Observed unemployment is 4.286% , or 0.197 percentage points above the fitted NAIRU. Smoothed states use the full updated sample; filtered states use observations available through each plotted quarter, with the same historical coefficients. The charts are model estimates, not direct measurements of potential output or the NAIRU.

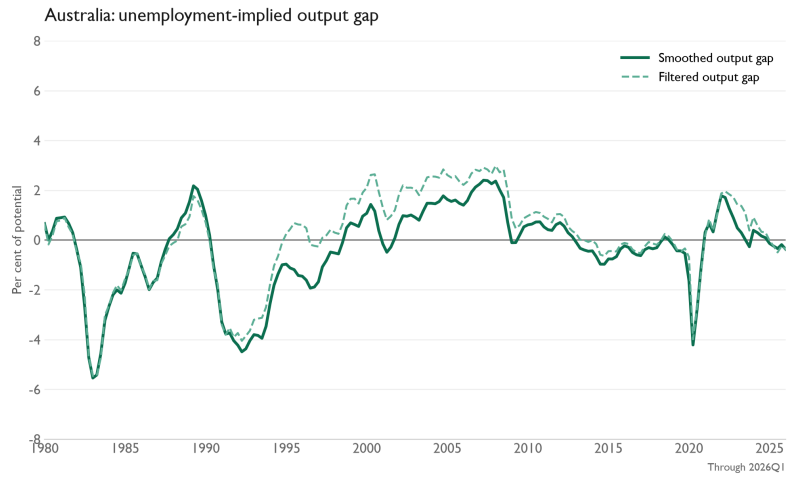


Figure 1: Output gap: smoothed and filtered estimates.

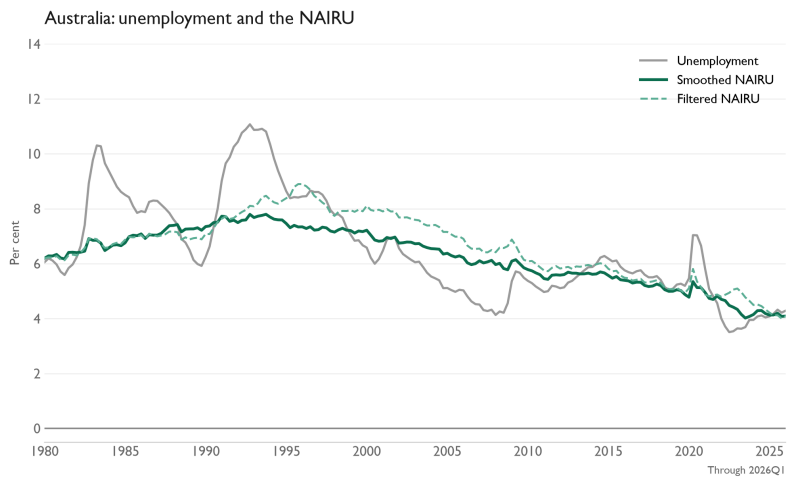


Figure 2: Observed unemployment and smoothed and filtered NAIRU.

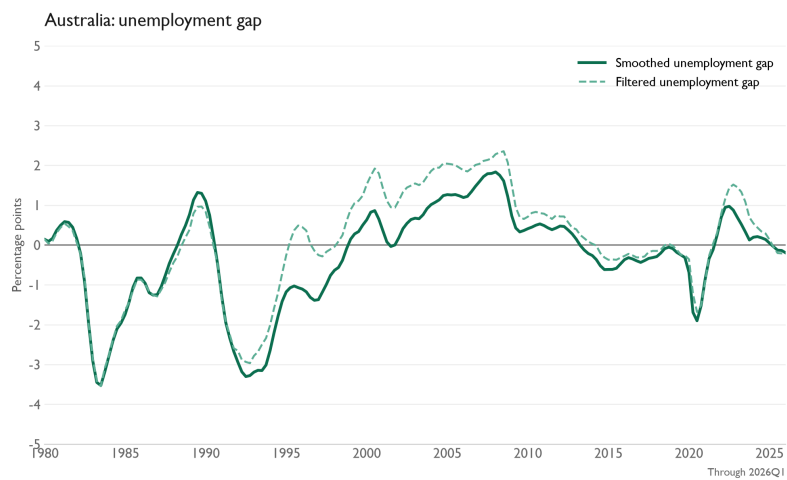


Figure 3: NAIRU minus observed unemployment: smoothed and filtered estimates.

3 Reading the current output gap

The negative gap and unemployment above the NAIRU are consistent with the negative unemployment-gap loading in this model. Yet the latest local inputs also show trimmed inflation of 3.466% and the model’s constructed expectation measure of 3.126% in 2026Q1. In the fitted inflation equation, the smoothed output gap contributes only -0.014 percentage points. The other fitted terms produce a conditional inflation value of 3.294% in total; actual inflation is 0.173 points higher. This is a decomposition using the smoothed state, not an out-of-sample forecast.

It is reasonable to question whether the model’s unemployment-driven gap captures every source of current price pressure. The observation does not, on its own, establish that the gap is too negative: lagged inflation, expectations, import prices, and the inflation residual also enter the measurement equation. The near-zero unemployment measurement-noise estimate and the NAIRU innovation floor make the latent gap sensitive to the imposed structure. Compare it with other capacity measures before treating its level as a definitive estimate.

4 State-space structure

The model uses seven states, $s_t = (g_t, g_{t-1}, g_{t-2}, y_t^*, y_{t-1}^*, u_t^*, u_{t-1}^*)'$, where y_t is log output, $g_t = y_t - y_t^*$ is a log output gap, and u_t^* is the latent structural unemployment rate. The reported output gap is $100g_t$, in percent of potential output. The transition equations are

$$g_t = \phi_1 g_{t-1} + \phi_2 g_{t-2} + \eta_t^g, \quad (1)$$

$$y_t^* = y_{t-1}^* + g^* + \eta_t^y, \quad (2)$$

$$u_t^* = u_{t-1}^* + \eta_t^u. \quad (3)$$

The output gap follows an AR(2), potential log output is a random walk with quarterly drift g^* , and the NAIRU is a random walk. The three state shocks are independent in the implemented diagonal covariance matrix. The state vector also carries lags needed by the observation equations.

At the beginning of each generation run, HP-based gap, potential output, and NAIRU values set a known initial state with zero prior covariance. The model recomputes these HP series from its historical workbook. The HP series initialize the filter and provide regression-based optimizer starting values; the saved maximum-likelihood coefficients are fitted by Python from the country’s own signals. This fixed initialization can matter for early likelihood terms and the resulting parameter estimates.

5 Australia: equations and estimates

Australia has three observed signals: log non-farm GDP y_t , unemployment u_t , and trimmed inflation π_t . Let π_t^e be the four-quarter moving average of year-on-year inflation, p_{t-1}^m the lagged four-quarter change in the import price deflator, and n_{t-1} lagged nominal unit labour cost growth. Let $D_t = 1$ after 1993Q1 and zero otherwise, and C_t denote the COVID dummy. The

implemented observation equations are

$$y_t = y_t^* + g_t + \phi_3 C_t + \varepsilon_t^y, \quad (4)$$

$$u_t = u_t^* + \lambda_3 g_t + \rho(u_{t-1} - u_{t-1}^*) + \varepsilon_t^u, \quad (5)$$

$$\begin{aligned} \pi_t = & D_t\{(1 - \beta_1 - \beta_2)\pi_t^e + \beta_1\pi_{t-1} + \beta_2\pi_{t-2}\} \\ & + (1 - D_t)\{(1 - \beta_1 - \beta_2 - \beta_3 - \gamma)\pi_t^e + \beta_1\pi_{t-1} + \beta_2\pi_{t-2} + \beta_3\pi_{t-3} + \gamma n_{t-1}\} \\ & + \lambda_1 g_t + \psi p_{t-1}^m + \varepsilon_t^\pi. \end{aligned} \quad (6)$$

The inflation expectation weight is therefore implied by the other coefficients, separately in the two regimes. It is not an independent parameter. Unit labour cost growth is a regressor in the pre-targeting Australian Phillips curve, not an additional observed signal.

Table 1: Australian maximum-likelihood coefficients. Values are rounded.

Parameter	Estimate	Meaning in the implemented model
β_1	-0.380964	Weight on one-quarter lagged inflation in both regimes.
β_2	-1.201930	Weight on two-quarter lagged inflation in both regimes.
β_3	-0.652679	Weight on three-quarter lagged inflation before targeting.
γ	0.004812	Weight on lagged unit labour cost growth before targeting.
λ_1	3.739261	Inflation response to a one-unit log output gap; a one percentage point gap gives about 0.0374 percentage points of inflation through this term.
λ_3	-29.930362	Unemployment response to the log output gap; a one percentage point gap corresponds to about -0.2993 percentage points of unemployment, holding other terms fixed.
ρ	0.626289	Persistence of lagged unemployment relative to the previous NAIRU in the unemployment equation.
ϕ_1	1.295685	First autoregressive coefficient of the latent output gap.
ϕ_2	-0.364960	Second autoregressive coefficient of the latent output gap.
ϕ_3	-0.000494	COVID dummy effect on observed log GDP, conditional on latent output.
ψ	0.025443	Loading of lagged four-quarter import-price change in inflation.
g^*	0.007482	Quarterly drift of potential log output, about 3.0% at an annualized log-growth rate.
ε_1	0.004307	Standard deviation of log GDP measurement noise; about 0.431% in log-output terms.
ε_2	0.000010	Standard deviation of unemployment measurement noise, in percentage points; at the imposed lower bound.
ε_3	0.334839	Standard deviation of inflation measurement noise, in percentage points.
ε_5	0.005644	Standard deviation of the log output-gap state shock; about 0.564% of potential output.
ε_6	0.004184	Standard deviation of the potential-log-output state shock.
ε_7	0.159770	Standard deviation of the quarterly NAIRU state shock, in unemployment percentage points.

The implied expectation weights are 2.5829 after 1993Q1 and 3.2308 before targeting. The large positive implied weights and negative lag weights are features of this fitted equation, not general statements about inflation dynamics. In particular, the near-zero fitted ε_2 makes the

unemployment equation almost exact conditional on the latent states. The code constrains ε_7 to be at least half the historical standard deviation of quarterly unemployment changes because the unrestricted likelihood allowed the NAIRU innovation scale to collapse. This bound and the ε_2 boundary solution mean the NAIRU and output-gap decomposition should be treated as sensitive to the model’s restrictions.

6 Likelihood fit and convergence

The script uses SciPy’s bounded L-BFGS-B optimizer on the Kalman-filter log likelihood. Starting guesses come from Australian historical data. The accepted-iteration history stores all 18 parameters, the full-sample log likelihood, and quarter-level contributions.

Fit statistic	Australia
Historical quarters	176
Accepted iterations	109
Initial log likelihood	234.1271
Iteration 25	549.1910
Iteration 50	549.2789
Final log likelihood	549.2792

Accepted steps increased the likelihood monotonically. The last history row agrees with the saved parameter JSON. Numerical convergence from these starting values and bounds does not establish a global optimum, tight identification, or forecast accuracy. The script does not export standard errors or out-of-sample forecast scores. The quarter-level likelihood file can identify observations that were hardest to predict: each contribution is a Kalman one-step-ahead predictive density term, not a squared-error metric. Early missing regressors cause the affected signals to be masked.

7 Reproduction and source files

From this model folder, run the generation script to re-estimate historical coefficients and overwrite the history CSVs. Then run the update script to apply the fixed fit to later quarters. Source files used for this report:

- Generation and update scripts in the **code** folder.
- Fitted parameter JSON at the model-folder root.
- The fit-history and likelihood-by-quarter CSVs in **output**.
- The updated state CSV in **results** and the **Update** tab of the local model-input workbook.

Numeric tables are a snapshot and do not update automatically when Python is rerun.